TASK 3: DRIVELINE NOISE AND VIBRATION DIAGNOSIS

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 7 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 91-106 | Months 24-28

Your task

Investigate a reported driveline noise or vibration on a heavy vehicle. Inspect the clutch/transmission/propshaft/final-drive path, use safe measurements and produce a justified repair plan.

What you must do

1. Clarify symptom, load/speed conditions, history and road-test authority.
2. Secure the vehicle and inspect mounts, shafts, joints, bearings and lubricant condition.
3. Measure free play, run-out, angles or other specified values.
4. Use findings to distinguish imbalance, misalignment, wear or control concerns.
5. Plan parts, time, risks, verification and customer/operator communication.

What good evidence looks like

SAFE CONTROL

Prevent engine start and driveline rotation unless an approved controlled test requires it; never work beneath an inadequately supported vehicle.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or driveline rig
- Manufacturer specifications and lubricant data
- Rated support and driveline measuring tools
- PPE, job card and repair-planning information

SOW ALIGNMENT

Block 7 - Transmission, Driveline, Hydraulics, Ancillaries and Complex Repair

KSB evidence focus: K8, K10, B2; reinforced K1-K4, K6, K11, S1-S6, S8-S10, B3

Task reference: HV_L3_Block_7_Task_3_Driveline_Noise_and_Vibration_Diagnosis

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Propshaft run-out		Run-out limit	
Joint / spline free play		Free-play limit	
Driveline angle		Angle specification	
Lubricant / fault finding		Root cause / repair plan	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why must driveline symptom conditions be recorded?

2. How can joint wear or angle error create vibration?

Knowledge check - continued

3. Explain controls before rotating or working near a driveline.

4. Why should repair planning include collateral damage and parts availability?

5. What checks verify a driveline repair?

Assessor/Workplace Mentor observation

- [ ] Confirms reproducible symptoms and safe test authority.
- [ ] Controls support, rotation and stored energy.
- [ ] Inspects and measures the driveline logically.
- [ ] Uses evidence to justify root cause.
- [ ] Plans repair, resources and verification clearly.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- [ ] PASS
- [ ] REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 7 | Task 3

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. Rear axle noisy

Checks: Confirm noise by load, speed and direction; check axle oil, bearings, backlash and final drive to vehicle specification.

Evidence: Record test conditions, measured play/noise and repair decision.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference:

2. Driveline heavy vibration

Checks: Confirm speed/load pattern; inspect propshaft phasing, joints, mounts, balance, run-out and alignment.

Evidence: Record measurements, source and safe post-repair verification.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

3. Differential lock inoperative

Checks: Check approved engagement conditions, request/interlocks, air or electrical actuator, feedback and mechanical linkage.

Evidence: Record command and response; verify engagement/disengagement only by approved method.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference:

4. Range change inoperative

Checks: Check complaint and operating conditions, selector request, supply/air pressure, actuator and gearbox feedback.

Evidence: Record inputs/outputs, root cause and approved functional check.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.